8.3: ATP powers cellular work by coupling exergonic reactions to endergonic reactions

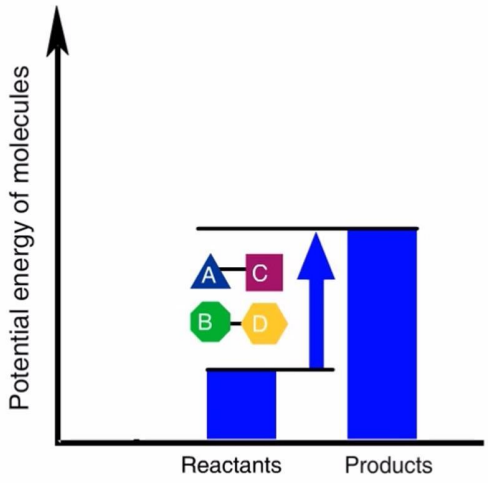
A cell does three main kinds of work:

- *Chemical work*, the pushing of endergonic reactions that would not occur spontaneously, such as the synthesis of polymers from monomers (chemical work will be discussed further here; examples are shown in Chapters 9 and 10)
- *Transport work*, the pumping of substances across membranes against the direction of spontaneous movement (see Concept 7.4)
- *Mechanical work*, such as the beating of cilia (see Concept 6.6), the contraction of muscle cells, and the movement of chromosomes during cellular reproduction

A key feature in the way cells manage their energy resources to do this work is **energy coupling**, the use of an exergonic process to drive an endergonic one. ATP is responsible for mediating most energy coupling in cells, and in most cases it acts as the immediate source of energy that powers cellular work.

Animation

Energy Coupling



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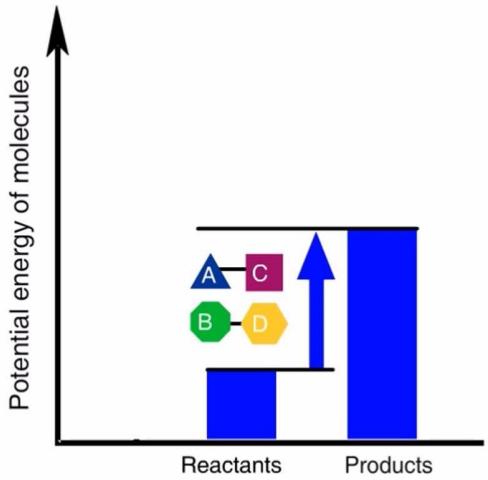

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The Structure and Hydrolysis of ATP

ATP **(adenosine triphosphate;**see Concept 4.3) contains the sugar ribose, with the nitrogenous base adenine and a chain of three phosphate groups (the triphosphate group) bonded to it **(Figure 8.9a)**. In addition to its role in energy coupling, ATP is also one of the nucleoside triphosphates used to make RNA (see Figure 5.23).

Figure 8.9

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The structure and hydrolysis of adenosine triphosphate (ATP).
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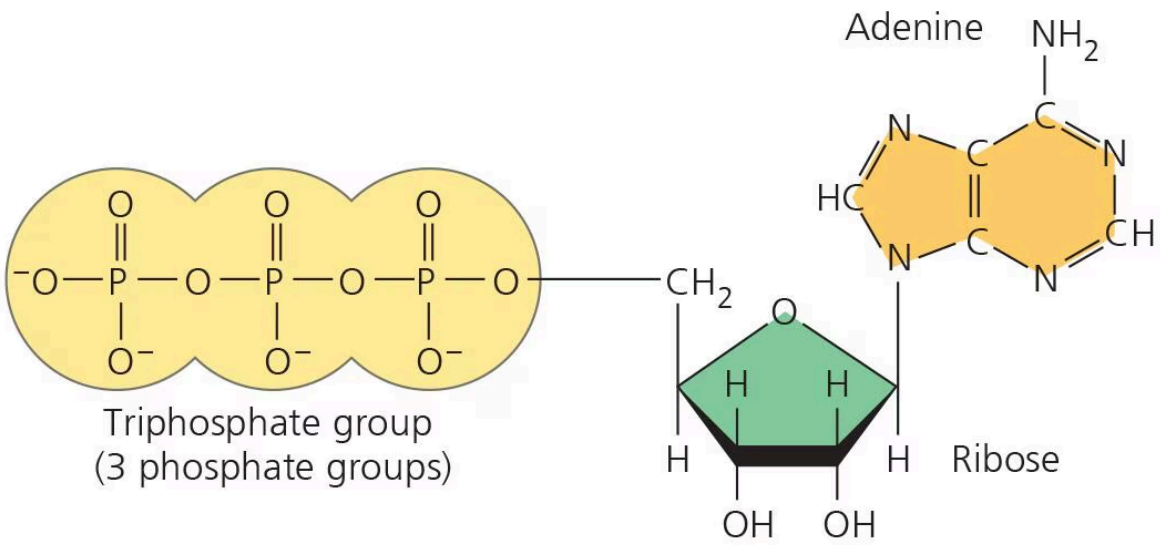
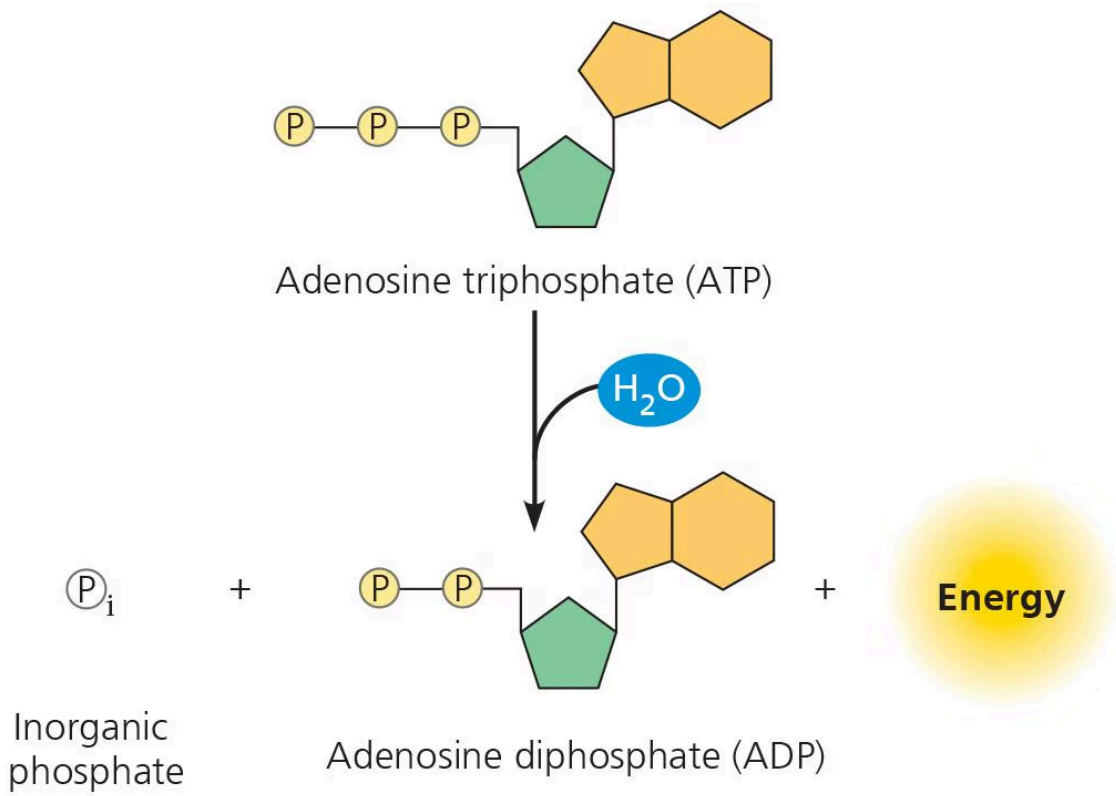


(a) The structure of ATP. In the cell, most hydroxyl groups of phosphates are ionized (—O^-).



(b) The hydrolysis of ATP. The reaction of ATP and water yields inorganic phosphate (P_i) and ADP and releases energy.

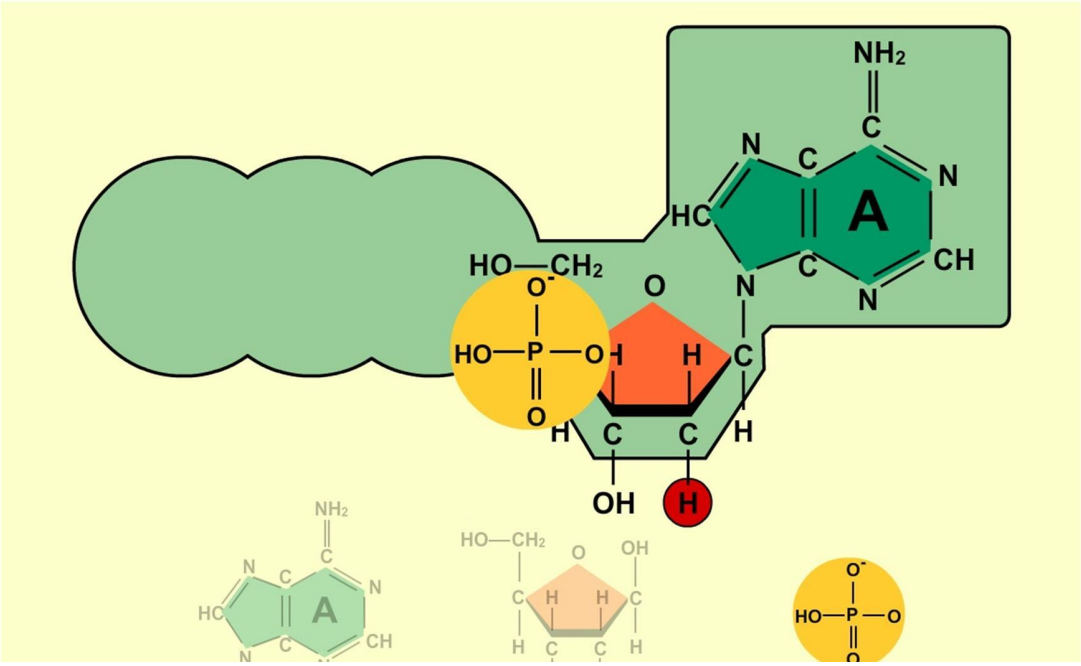
*Throughout this book, the chemical structure of the triphosphate group seen in **(a)** will be represented by the three joined yellow circles shown in **(b)**.*

Part a. The structure of ATP: triphosphate group attached to a ribose sugar, which is attached to adenine. In the cell, most hydroxyl groups of phosphates are ionized (O minus) A triphosphate group consists of three phosphate groups (each a central P sharing a double bond with one O atom and single bonded to 3 other O atoms. In the triphosphate, P's are bonded to O's in the adjacent phosphate group with a single bond. The triphosphate group is attached to a ribose sugar, which consists of a 5-carbon ring. The ribose sugar is attached to an adenine, which is a nitrogenous (nitrogen containing) base consisting of a double-ring structure.

Part b. The reaction of A T P and water yields inorganic phosphate (circle P sub i) and A D P and releases energy. A T P, symbolized here with three yellow circles (phosphate groups), a green pentagon (ribose sugar) and gold double rings (adenine) is hydrolyzed by water, forming inorganic phosphate (circle P sub i), A D P (two phosphate groups, sugar, and adenine) and energy is released.

Animation

The Structure of ATP



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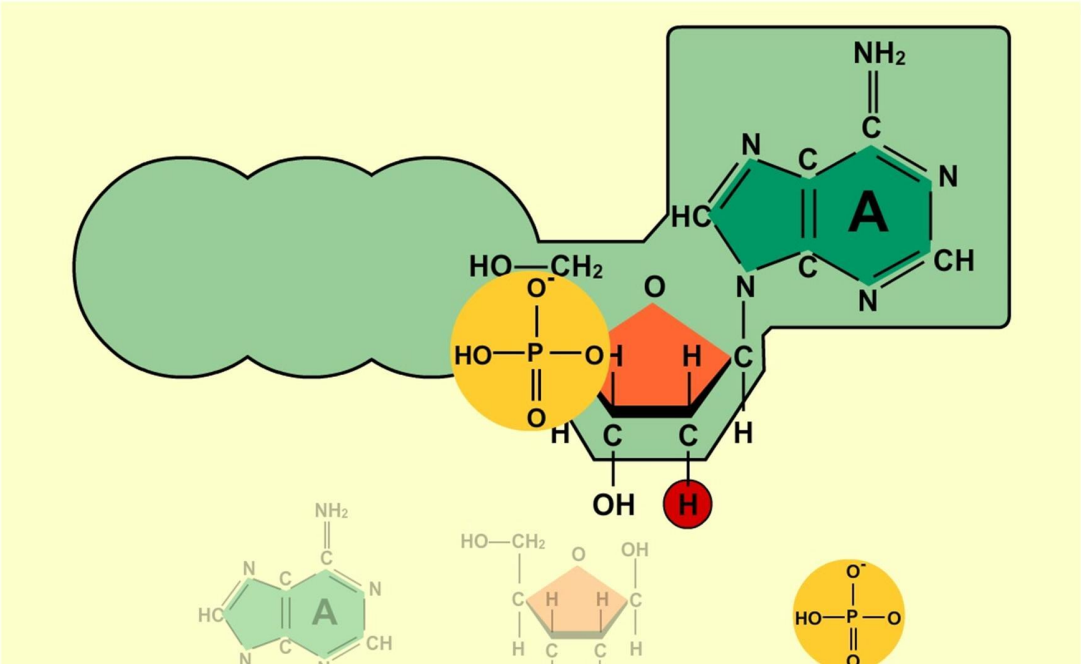

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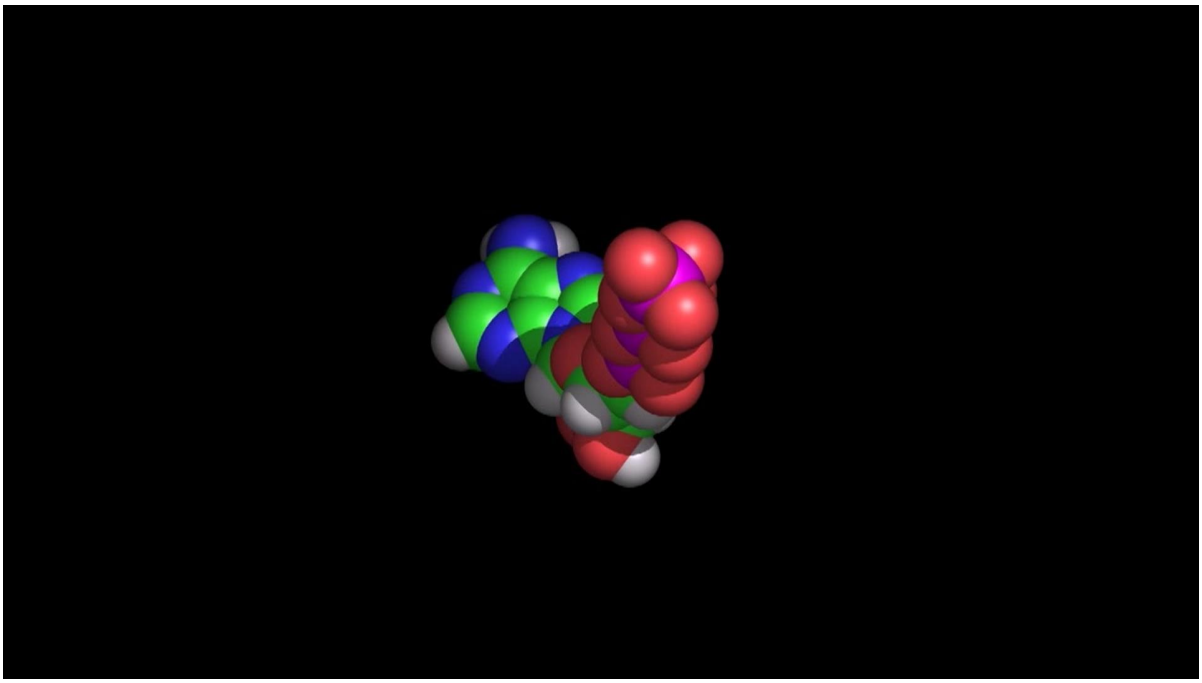
Animation

Space-Filling Model of ATP



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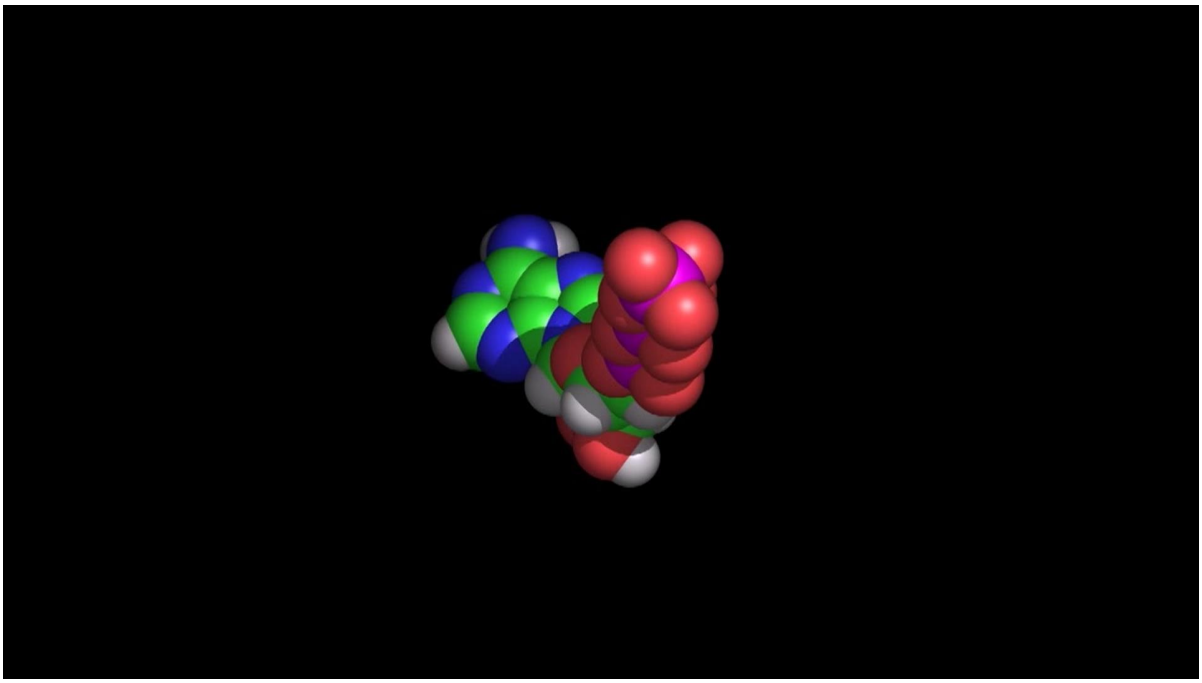

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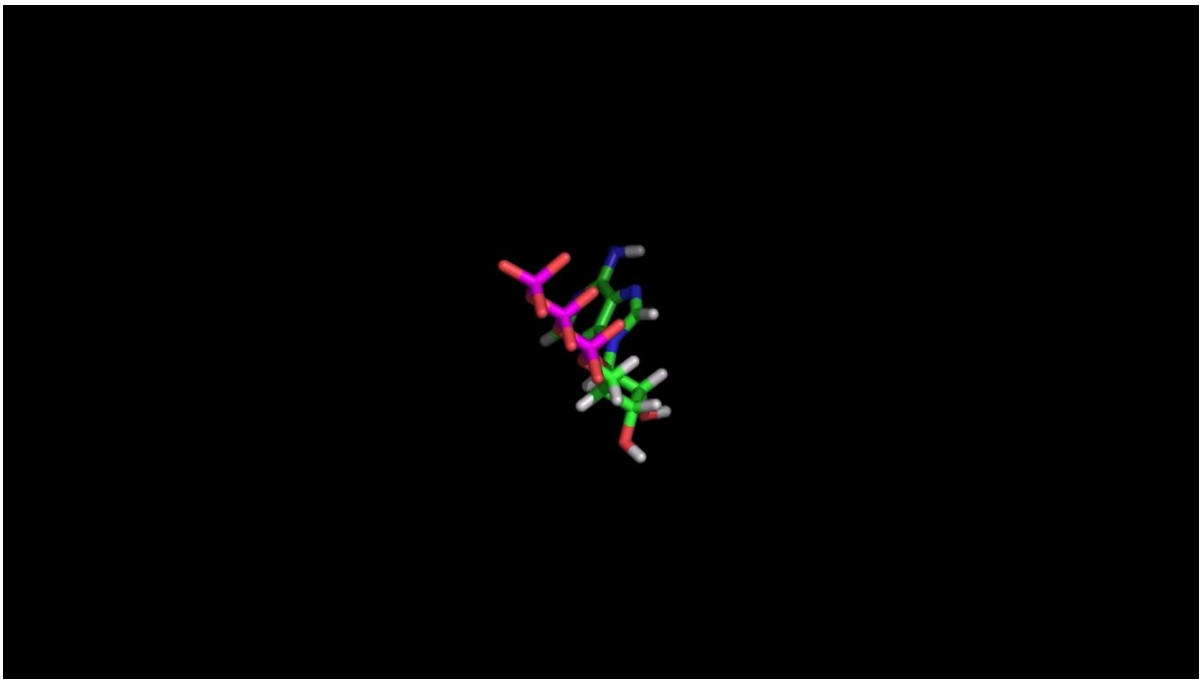
Animation

Stick Model of ATP



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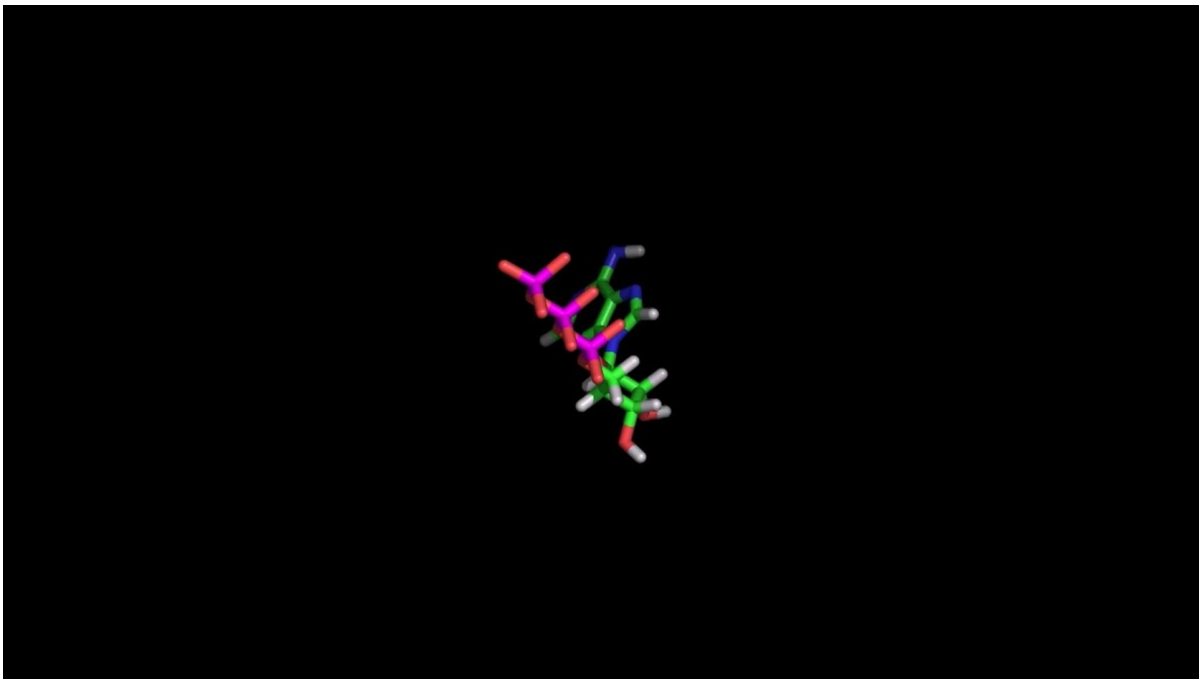

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The bonds between the phosphate groups of ATP can be broken by hydrolysis. When the terminal phosphate bond is broken by addition of a water molecule, a molecule of inorganic phosphate ($HOPO_3^{2-}$, abbreviated as Ⓟi throughout this book) leaves the ATP, which becomes adenosine diphosphate, or ADP **(Figure 8.9b)**. The reaction is exergonic and releases 7.3 kcal of energy per mole of ATP hydrolyzed:

$ATP + H_2O \rightarrow ADP + Ⓟi \quad \Delta G = -7.3$ kcal/mol (−30.5 kJ/mol)

This is the free-energy change measured under standard conditions. In the cell, conditions do not conform to standard conditions, primarily because reactant and product concentrations differ from 1M. For example, when ATP hydrolysis occurs under typical cellular conditions, the actual ΔG is about −13 kcal/mol,78% greater than the energy released by ATP hydrolysis under standard conditions.

Because their hydrolysis releases energy, the phosphate bonds of ATP are sometimes referred to as high-energy phosphate bonds, but the term is misleading. The phosphate bonds of ATP are not unusually strong bonds, as “high-energy” may imply; rather, the reactants (ATP and water) themselves have high energy relative to the energy of the products (ADP and Ⓟi). The release of energy during the hydrolysis of ATP comes from the chemical change of the system to a state of lower free energy, not from the phosphate bonds themselves.

ATP is useful to the cell because the energy it releases on losing a phosphate group is somewhat greater than the energy most other molecules could deliver. But why does this hydrolysis release so much energy? If we reexamine the ATP molecule in Figure 8.9a, we can see that all three phosphate groups are negatively

charged. These like charges are crowded together, and their mutual repulsion contributes to the instability of this region of the ATP molecule. The triphosphate tail of ATP is the chemical equivalent of a compressed spring.

How ATP Provides Energy That Performs Work

When ATP is hydrolyzed in a test tube, the release of free energy merely heats the surrounding water. In an organism, this same generation of heat can sometimes be beneficial. For instance, the process of shivering uses ATP hydrolysis during muscle contraction to warm the body. In most cases in the cell, however, the generation of heat alone would be an inefficient (and potentially dangerous) use of a valuable energy resource. Instead, the cell's proteins harness the energy released during ATP hydrolysis in several ways to perform the three types of cellular work—chemical, transport, and mechanical.

For example, with the help of specific enzymes, the cell is able to use the high free energy of ATP to drive chemical reactions that, by themselves, are endergonic. If the ΔG of an endergonic reaction is less than the amount of energy released by ATP hydrolysis, then the two reactions can be coupled so that, overall, the coupled reactions are exergonic. This usually involves phosphorylation, the transfer of a phosphate group from ATP to some other molecule, such as the reactant. The recipient molecule with the phosphate group covalently bonded to it is then called a **phosphorylated intermediate**. The key to coupling exergonic and endergonic reactions is the formation of this phosphorylated intermediate, which is more reactive (less stable, with more free energy) than the original unphosphorylated molecule **(Figure 8.10)**.

Figure 8.10

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How ATP drives chemical work: energy coupling using ATP hydrolysis.
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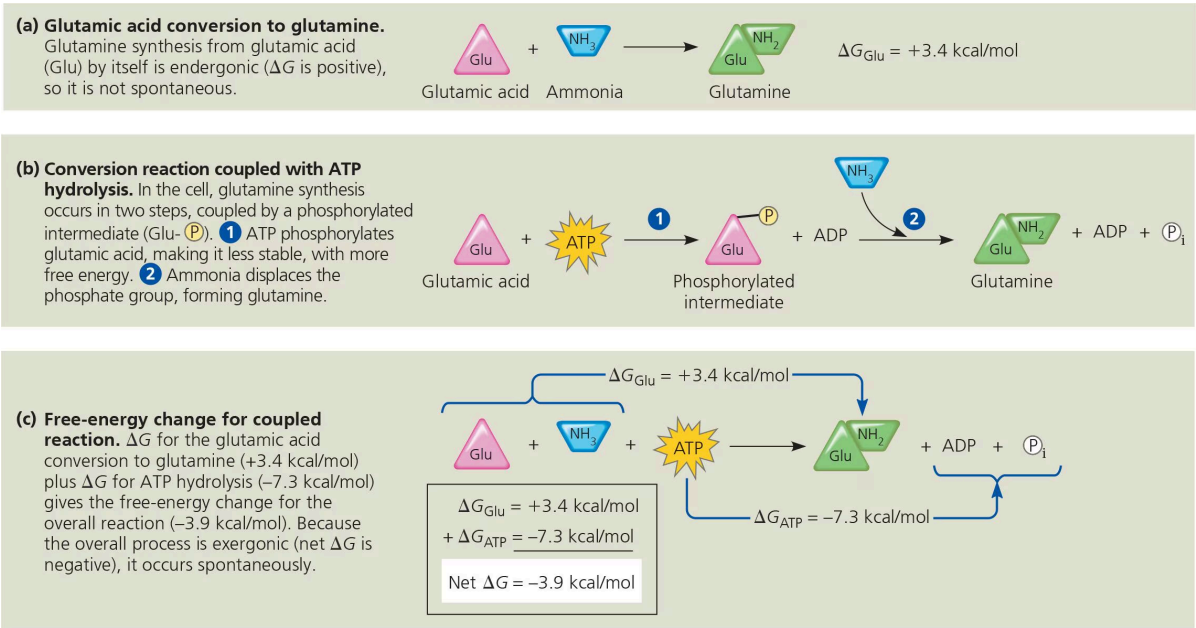


In this example, the exergonic process of ATP hydrolysis drives an endergonic process—synthesis of the amino acid glutamine.

Part a. Glutamic acid conversion to glutamine. Glutamine synthesis from glutamic acid (Glu) by itself is endergonic (Delta G is positive), so it is not spontaneous. Glutamic acid (Glu) plus Ammonia (N H 3) goes to Glutamine (Glu bonded to N H 2). Delta G sub Glu = +3.4 k cal per mole.

Part b. Conversion reaction coupled with A T P hydrolysis. In the cell, glutamine synthesis occurs in two steps, coupled by a phosphorylated intermediate. In the cell, glutamine synthesis occurs in two steps, coupled by a phosphorylated intermediate. A T P phosphorylates glutamic acid, making it less stable, with more free energy. Ammonia displaces the phosphate group, forming glutamine.

- Step 1: Glutamic acid (Glu) plus A T P goes to Phosphorylated intermediate (Glu with phosphate group covalently bonded to it) plus A D P;
- Step 2: Ammonia (N H 3) is added and Glutamine (Glu N H 3) + A D P + phosphorylated intermediate (circle P sub i) is formed.

Part c. Free-energy change for coupled reaction. Delta G for the glutamic acid conversion to glutamine plus Delta G for A T P hydrolysis gives the free-energy change for the overall reaction. Delta G for the glutamic acid conversion to glutamine (+3.4 kcal/mol) plus Delta G for ATP hydrolysis (negative 7.3 k cal per mole) gives the free-energy change for the overall reaction (negative 3.9 k cal per mol). Because the overall process is exergonic (net Delta G is negative), it occurs spontaneously. Glu + N H 3 + A T P goes to Glu N H 2 + A D P + Inorganic phosphate. Delta G Glu = +3.4 k cal per mol which results from the glutamic acid conversion to glutamine; Delta G sub A T P = negative 7.3 k cal per mol which results from the A T P conversion to A D P + inorganic phosphate; Net Delta G = negative 3.9 k cal per mol.

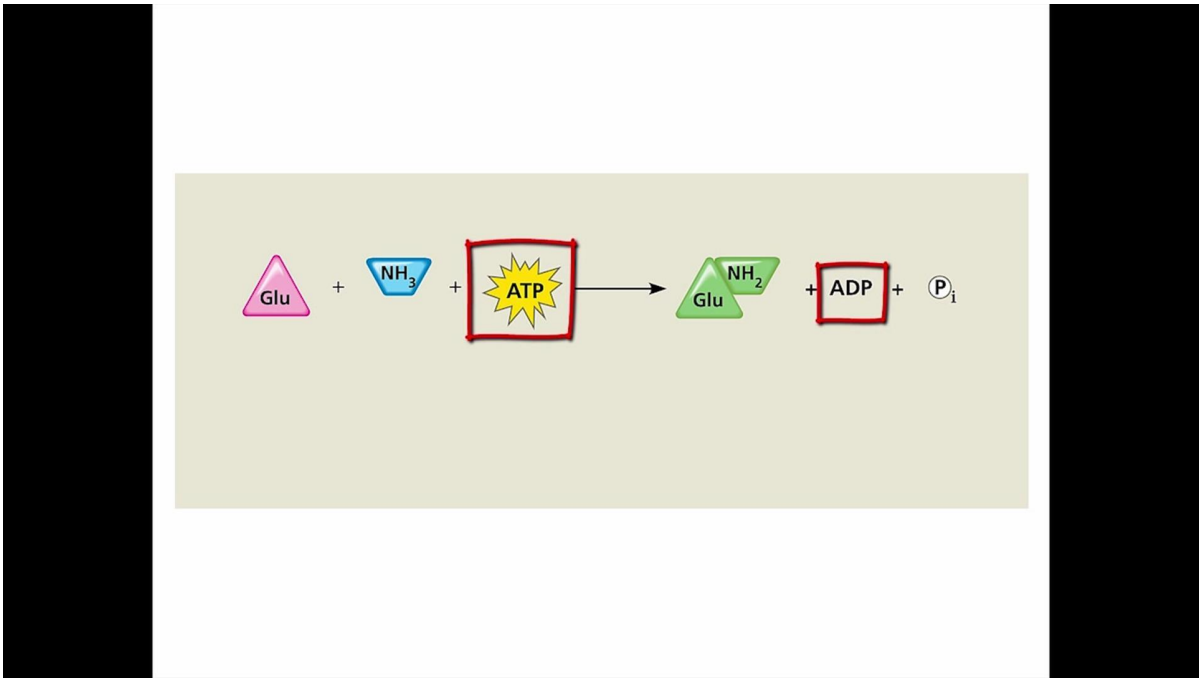
MAKE CONNECTIONS *Referring to*Figure 5.14*, explain why glutamine (Gln) is drawn in this figure as a glutamic acid (Glu) with an amino group attached.*

*For suggested answer, see*Appendix A.*

Figure 8.10 Walkthrough

How ATP drives chemical work: energy coupling using ATP hydrolysis

Watch an explanation of this figure in the Figure 8.10 Walkthrough: How ATP drives chemical work: energy coupling using ATP hydrolysis.



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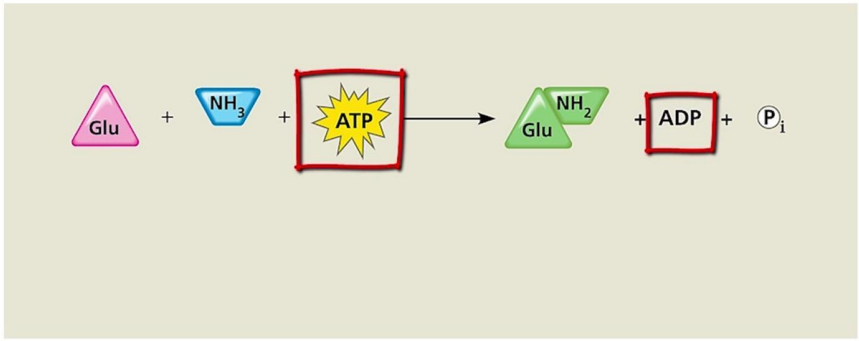

Transport and mechanical work in the cell are also nearly always powered by the hydrolysis of ATP. In these cases, ATP hydrolysis leads to a change in a protein's shape and often its ability to bind another molecule. Sometimes this occurs via a phosphorylated intermediate, as seen for the transport protein in **Figure 8.11a**. In most instances of mechanical work involving motor proteins "walking" along cytoskeletal elements **(Figure 8.11b)**, a cycle occurs in which ATP is first bound noncovalently to the motor protein. Next, ATP is hydrolyzed, releasing ADP and Ⓟi. Another ATP molecule can then bind. At each stage, the motor protein changes its shape and ability to bind the cytoskeleton, resulting in movement of the protein along the cytoskeletal track. Phosphorylation and dephosphorylation promote crucial protein shape changes during many other important cellular processes as well.

Figure 8.11

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How ATP drives transport and mechanical work.
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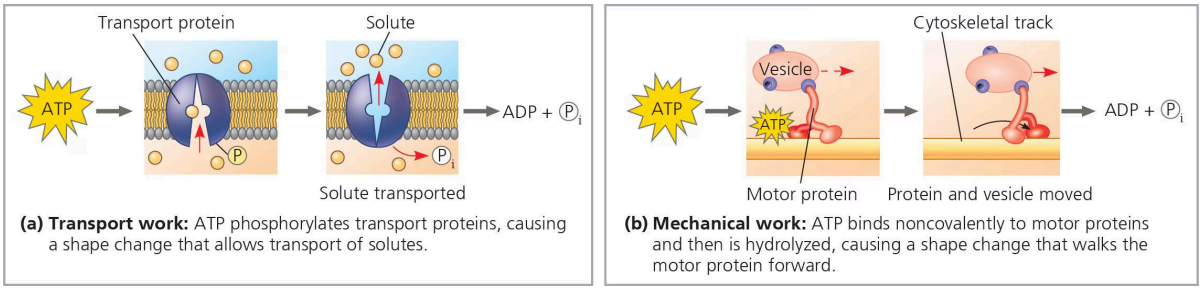

(a) Transport work: ATP phosphorylates transport proteins, causing a shape change that allows transport of solutes.

(b) Mechanical work: ATP binds noncovalently to motor proteins and then is hydrolyzed, causing a shape change that walks the motor protein forward.

Part a. Transport work: A T P phosphorylates transport proteins, causing a shape change that allows transport of solutes. A T P hydrolysis leads to a change in the shape of the transport protein allowing it to transport a solute. A D P plus inorganic phosphate is released.

Part b. Mechanical work: A T P binds noncovalently to motor proteins and then is hydrolyzed, causing a shape change that walks the motor protein forward. A T P is bound noncovalently to the motor protein; the motor protein and vesicle "walk" along the cytoskeletal track and then A D P and inorganic phosphate are released.

ATP hydrolysis causes changes in the shapes and binding affinities of proteins. This can occur either **(a)** directly, by phosphorylation, as shown for a membrane protein carrying out active transport of a solute (see also Figure 7. 16 and the proton pump in Figure 6. 32, upper left) , or **(b)** indirectly, via noncovalent binding of ATP and its hydrolytic products, as is the case for motor proteins that move vesicles (and other organelles) along cytoskeletal “tracks” in the cel l (see also Figures 6. 21 and 6. 32, lower right).

The Regeneration of ATP

An organism at work uses ATP continuously, but ATP is a renewable resource that can be regenerated by the addition of phosphate to ADP **(Figure 8.12)**. The free energy required to phosphorylate ADP comes from exergonic breakdown reactions (catabolism) in the cell. This shuttling of inorganic phosphate and energy is called the ATP cycle, and it couples the cell’s energy-yielding (exergonic) processes to the energy-consuming (endergonic) ones. The ATP cycle proceeds at an astonishing pace. For example, a working muscle cell recycles its entire pool of ATP in less than a minute. That turnover represents 10 million molecules of ATP consumed and regenerated per second per cell. If ATP could not be regenerated by the phosphorylation of ADP, humans would use up nearly their body weight in ATP each day.

Figure 8.12

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The ATP cycle.
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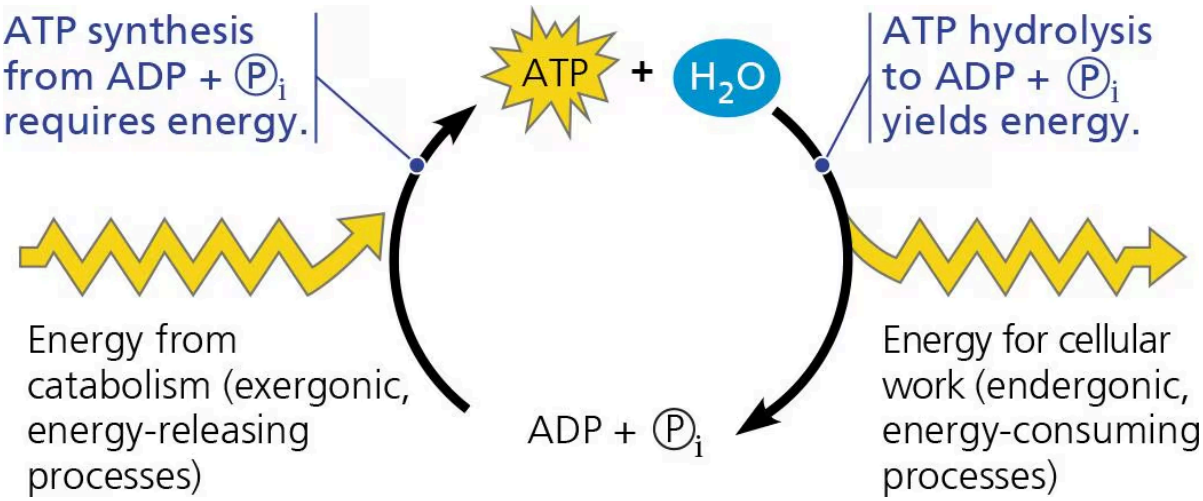
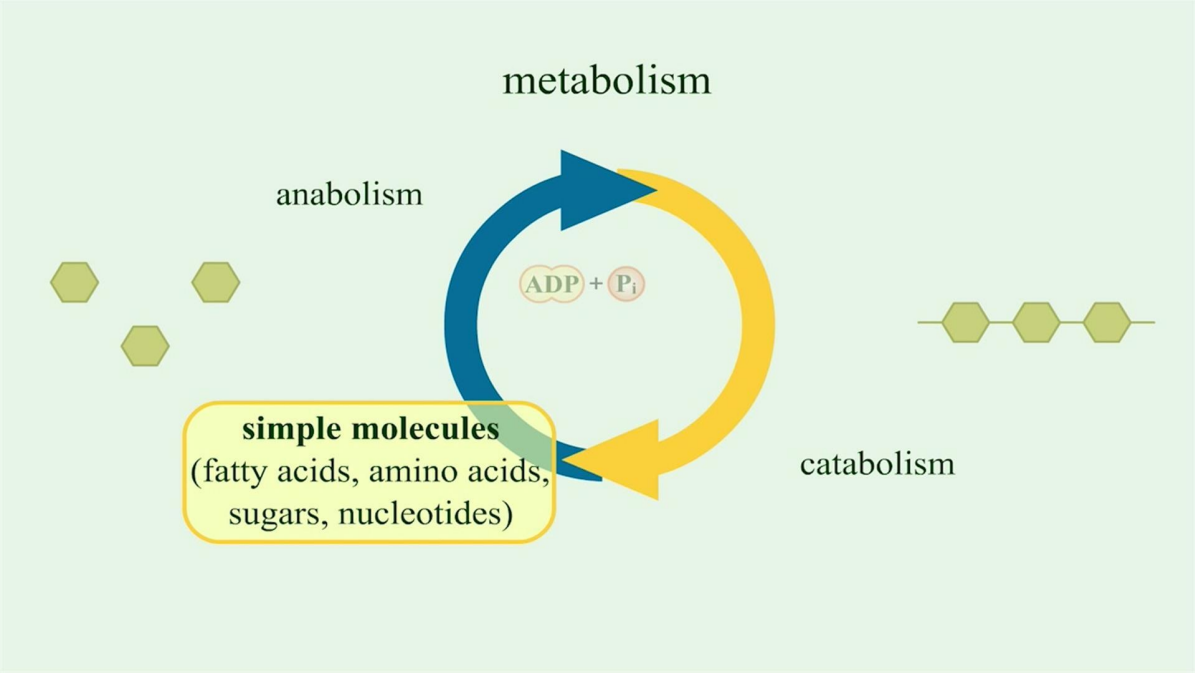


Energy released by breakdown reactions (catabolism) in the cell is used to phosphorylate ADP, regenerating ATP. Chemical potential energy stored in ATP drives most cellular work.

A T P synthesis from A D P plus inorganic phosphate requires energy which is supplied from catabolism (exergonic, energy-releasing processes). A T P hydrolysis to ADP plus inorganic phosphate and energy from cellular work (endergonic, energy-consuming processes) is yielded.

Animation

Metabolism Overview



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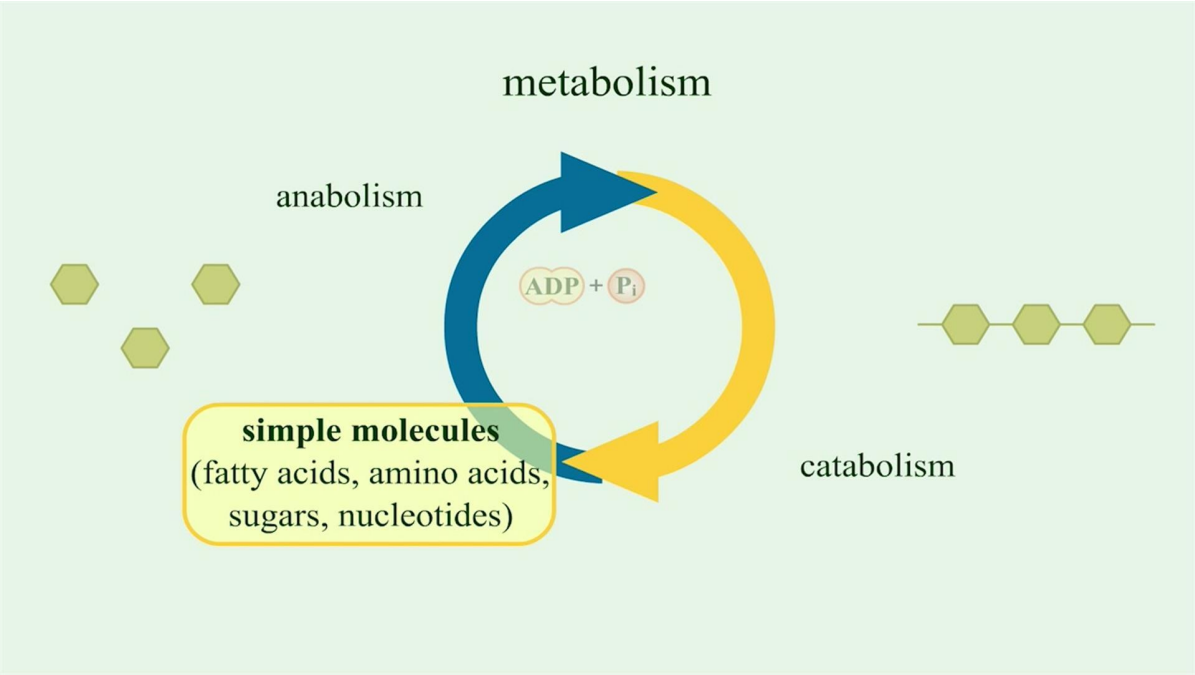

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Because both directions of a reversible process cannot be downhill, the regeneration of ATP from ADP and Ⓟi is necessarily endergonic:

ADP+Ⓟi→ATP+H2OΔG=+7.3 kcal/mol (+30.5 kJ/mol)(standard conditions)

Since ATP formation from ADP and Ⓟi is not spontaneous, free energy must be spent to make it occur. Catabolic (exergonic) pathways, especially cellular respiration, provide the energy for the endergonic process of making ATP. Plants also use light energy to produce ATP. Thus, the ATP cycle is a key player in bioenergetics, functioning as a revolving door through which energy passes during its transfer from catabolic to anabolic pathways.